

WASP-12b

R-2357

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_141222 · created 2026-08-19T22:59:36+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457013.78597 +/- 0.00012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1467 +/- 0.0073
Transit depth (Rp/Rs)^2	2.15 +/- 0.21 [%]
Orbital Inclination (inc)	80.57 +/- 2.34
Airmass coefficient 1 (a1)	1.077 +/- 0.011
Airmass coefficient 2 (a2)	-0.069 +/- 0.015
Scatter in the residuals of the lightcurve fit is	0.98 %
Best Comparison Star	#2 - [281, 289]
Optimal Aperture	3.98
Optimal Annulus	6.79
Transit Duration (day)	0.1211 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1467 $\pm$ 0.0073 vs 0.1178 (deviation 3.72 σ)
Duration fit vs published	0.1211 d vs 0.125 d (deviation 0.55 σ)
Mid-time O-C	-0.3 min
Depth SNR	18.9
Residual scatter	0.98 %

Light curve

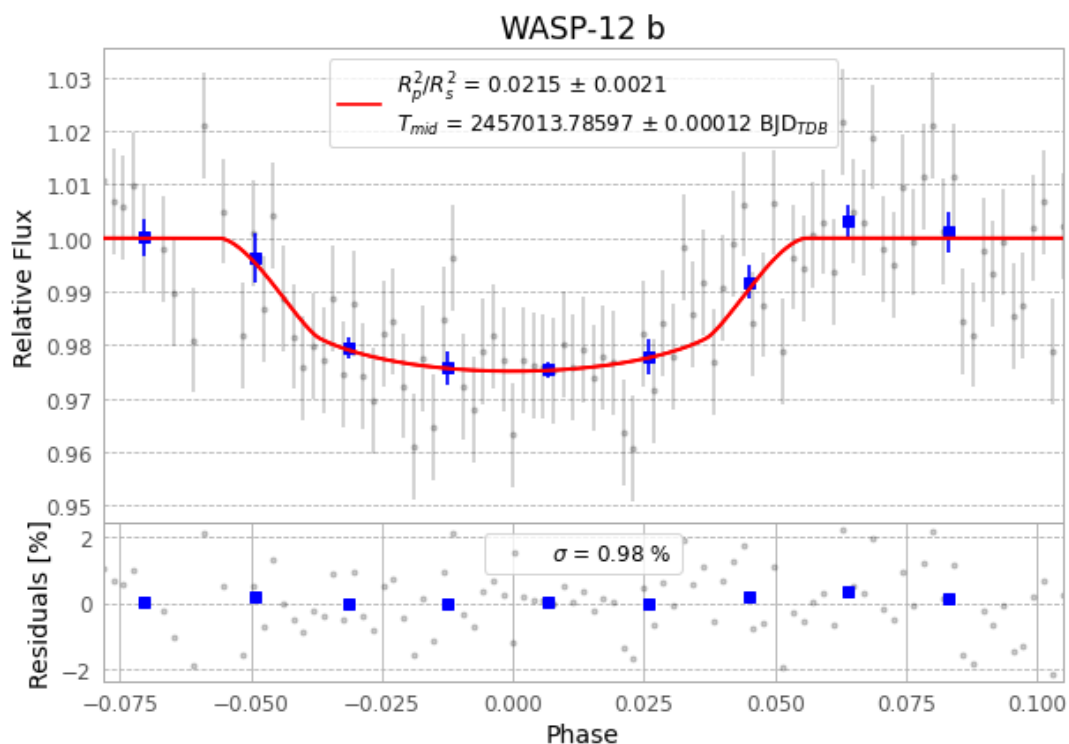


Figure 1 — FinalLightCurve_WASP-12 b_2014-12-21.png (SHA-256 b871841ec8f9a6f7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [360, 252], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8edabc2cbf7e6e0c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

100 FITS frames (66 MB), WASP-12141222043012.FITS → WASP-12141222092712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-141222.log (d46c34d0b270...), FinalLightCurve_WASP-12 b_2014-12-21.png (b871841ec8f9...), FinalLightCurve_WASP-12 b_2014-12-21.csv (52a80c8e4170...)

Compute resources

Wall time 132.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 22:59Z.